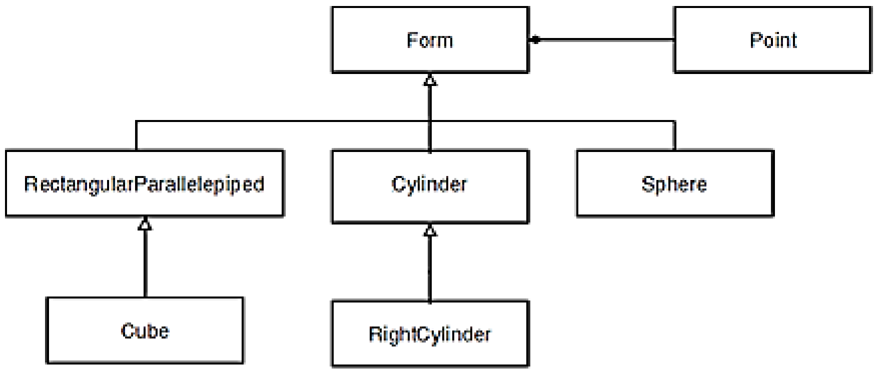
Holy Spirit University of Kaslik
Faculty of Engineering

GIN231 - Data Structures and Algorithms
Project #2

Implement in Java the classes defined in the following UML diagram:



Requirements:

- Classes must be in the same package.
- Lengths are in meters (m).
- Angles are in radians (rd).
- A cylinder is characterized by its density, its center of gravity, the radius of its base circles, the distance between the centers of its two base circles, and the angle between the axis of the cylinder and the base (see appendix 2).
- A right cylinder is a cylinder having an angle between the axis of the cylinder and the base.
- A sphere is characterized by its density, center of gravity, and its radius.
- A rectangular parallelepiped is characterized by its density, its center of gravity, its width, its length, and its height.
- A cube is a rectangular parallelepiped such that: width = length = height.
- The center of gravity is an object of the class Point (see appendix 1).
- All forms must have the following methods:
    - Constructor with no arguments.
    - Constructor with initial state.
    - Copy constructor.
    - Getters and Setters.
    - An override of the method `equals`.
    - An override of the method `toString`.
    - `getVolume`: which returns the volume of the current form (see annex 2).
    - `getWeight`: which returns the weight of the current form.
    - `haveSameVolume`: class method that returns a `boolean` to indicate if the 2 forms received as arguments have the same volume.
    - `hasSameVolumeAs`: method that returns a `boolean` to indicate if the current object has the same volume as the form received as argument.
    - `isBiggerThan`: method that returns a `boolean` to indicate if the current object is larger than the form received as argument.
    - `isSmallerThan`: method that returns a `boolean` to indicate if the current object is smaller than the form received as argument.
    - `haveSameWeight`: class method that returns a `boolean` to indicate if the 2 forms received as arguments have the same weight.

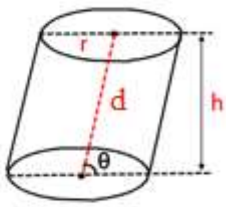
- `hasSameWeightAs`: method that returns a `boolean` to indicate if the current object has the same weight as the form received as argument.
- `isHeavierThan`: method that returns a `boolean` to indicate if the current object is heavier than the form received as argument.
- `isLighterThan`: method that returns a `boolean` to indicate if the current object is lighter than the form received as argument.
- `moveTo`: method that moves the current object by moving its center of gravity to the point received as argument.

- You should also provide a test program.

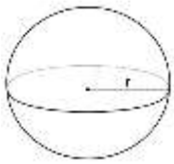
<u>Annex 1: class Point</u>

Point
- x : double - y : double - z : double
+ Point() + Point(double, double, double) + Point(Point) + equals (Object) : boolean + getX () : double + getY () : double + getZ () : double + toString() : String + moveTo(Point)

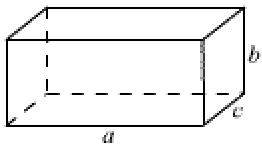
<u>Annex 2:</u>



$V = \pi r^2 h$



$V = 4/3\, \pi r^3$



$V = a \times b \times c$

weight = density × volume × g ; where g = 9.81 m/s² ($g = 9.81\ m/s^2$)